

A disproof of Conjecture 00000002221

$\mathbb{F}_2^3$ has Krull dimension 0 but a Stone space with $3 > 2^{2^0}$ points

1 The conjecture and its readings

Conjecture 00000002221 reads: *Definition: The constructible topology on $\text{Spec } R$. Conjecture: The cardinality of the Stone space of the Booleanization (complemented-closure algebra) of the spectrum is at most 2^{2^d} with $d = \dim$, explicitly (Booleanization cardinality).*

Here R is a commutative ring, $d = \dim R$ is its Krull dimension (equal to the Krull dimension of the poset $\text{Spec } R$), and the Stone space of a Boolean algebra B is the set of Boolean homomorphisms $B \rightarrow \mathbf{2} = \{0, 1\}$ (equivalently, the ultrafilters of B).

The phrase “Booleanization (complemented-closure algebra) of the spectrum” is not a standard term. We refute the following readings of it.

- (R1) B is the Boolean algebra of *constructible* subsets of $\text{Spec } R$: the Boolean subalgebra of the power set generated by the retrocompact open sets (Mathlib’s `Topology.IsConstructible`). This is the reading suggested by the “Definition” line, because the Stone space of this algebra is $\text{Spec } R$ with the constructible topology.
- (R2) B is the Boolean algebra of *clopen* subsets of $\text{Spec } R$, i.e. the complemented elements of the lattice of open sets.
- (R3) The “Stone space” is taken to be $\text{Spec } R$ itself with the constructible topology.
- (R4) B is any Boolean algebra with a homomorphism of bounded lattices $\varphi : B \rightarrow \mathcal{P}(\text{Spec } R)$ whose image contains every singleton. This includes the full power set.

For the ring used below, every subset of $\text{Spec } R$ is both clopen and constructible (Lemma 2). So the algebras in (R1) and (R2) are the full power set, and both are instances of (R4). Readings in which B is not a Boolean algebra of subsets of $\text{Spec } R$ are not treated.

2 The counterexample

Let K be a field, $N \geq 1$, and $R = K^N = \prod_{i < N} K$. For $i < N$ let $\mathfrak{p}_i = \ker(\text{ev}_i)$, where $\text{ev}_i : R \rightarrow K$ is the i -th projection.

Lemma 1. $\dim R = 0$.

Proof. R is a finite-dimensional K -algebra, hence Artinian, so every prime ideal is maximal. Since $R \neq 0$, $\text{Spec } R \neq \emptyset$, and therefore $\dim R = 0$. $\square$

Lemma 2. *The map $\coprod_{i < N} \text{Spec } K \rightarrow \text{Spec } R$, $(i, \mathfrak{q}) \mapsto \text{ev}_i^{-1}(\mathfrak{q})$, is a homeomorphism. Consequently:*

- $\text{Spec } R$ is a finite discrete space;
- the primes $\mathfrak{p}_0, \dots, \mathfrak{p}_{N-1}$ are pairwise distinct;
- every subset of $\text{Spec } R$ is clopen and constructible.

Proof. The homeomorphism is Mathlib’s `PrimeSpectrum.sigmaToPiHomeo`, valid for finite products. $\text{Spec } K$ is a single point, so the coproduct is discrete and finite, and the $\mathfrak{p}_i$ are its distinct images. In a discrete space every set is open and closed. In a finite space every set is compact, so every open set is retrocompact. Hence every set is a retrocompact open, and in particular constructible. $\square$

Lemma 3. *Let X be a set, B a Boolean algebra, and $\varphi : B \rightarrow \mathcal{P}(X)$ a homomorphism of bounded lattices whose image contains every singleton. Then $x \mapsto (b \mapsto [x \in \varphi(b)])$ is an injective map from X into the Stone space $\text{Hom}(B, \mathbf{2})$. Hence $|\text{Hom}(B, \mathbf{2})| \geq |X|$.*

Proof. Evaluation at a point preserves $\cup, \cap, \emptyset, X$, so each map is a Boolean homomorphism. If $x \neq y$, choose b with $\varphi(b) = \{x\}$; the two homomorphisms differ at b . $\square$

Theorem 4. *For every field K and every $N \geq 1$, $\dim K^N = 0$. In each of the readings (R1)–(R4), the Stone space of $\text{Spec } K^N$ has at least N points. In particular, for $R = \mathbb{F}_2^3$ we have $d = 0$ and the bound would be $2^{2^0} = 2$, but the Stone space has at least 3 points. So the conjecture is false in readings (R1), (R2) and (R3), and in (R4) for every admissible B .*

Remark 5. In fact $\text{Spec } K^N$ has exactly N points, and the Stone space of the full power set of an N -point set has exactly N points. Only the lower bound is needed and formalized. Even if “cardinality” were applied to the Boolean algebra itself ($2^3 = 8$ elements), the bound 2 would fail.

3 Relation to the earlier submission

PR #219 by orionsheep was closed unmerged. The reason was that its Lean “proves only $2^{(2^0)} = 2$, $3 > 2$, $8 > 2$; the ring k^3 , Spec, Booleanization, Stone space, and Krull dimension appear nowhere.” Its prose also counted the Stone space as having $2^3 = 8$ points. That is the cardinality of the Boolean algebra $\mathcal{P}(\text{Spec } k^3)$, not of its Stone space, which has 3 points. The conclusion survives, since $3 > 2$ as well.

The present Lean file works with the actual objects:

- the ring `Fin N → K` and its Krull dimension `ringKrullDim`;
- `PrimeSpectrum` with its Zariski topology;
- the constructible Boolean algebra, as Mathlib’s `BooleanSubalgebra.closure` of the retrocompact opens;
- the clopen algebra `TopologicalSpace.Clopens`;
- Mathlib’s `WithConstructibleTopology`;
- the Stone space `BoundedLatticeHom B Bool`.

The main theorems quantify over all commutative rings R and refute the bound for each reading.

4 Lean formalization

The file is `lean/Conjecture2221/Basic.lean` (namespace `C2221`). It imports only `Mathlib`.

- Definitions:
 - `StoneSpace B := BoundedLatticeHom B Bool`;
 - `constructibleAlgebra R := BooleanSubalgebra.closure {U | IsOpen U ∧ IsRetrocompact U}` (membership is `IsConstructible`, see `mem_constructibleAlgebra_iff`);
 - `clopensHom`, the inclusion of clopens into sets;
 - `evalAt`, the evaluation ultrafilters $b \mapsto [x \in \varphi(b)]$ (principal for the finite power-set algebras of `R1` and `R2`; for a general `B` in `R4` they need not be principal, and only evaluation and separation are used);
 - `bound d := 2^2^d`.
- `evalAt_injective` and `mk_le_stone`: Lemma 3.
- `ringKrullDim_pi : ringKrullDim (Fin N → K) = 0` for $N \neq 0$.
- Facts about `Spec`:
 - instances `discrete_spec` and `finite_spec`;
 - `pt_injective` and `card_spec`: $N \leq \# \text{Spec}$;
 - `every_subset`: every subset is `IsClopen` and `IsConstructible`.
- Lower bounds $N \leq \#$ for each reading:
 - `stone_card_of_hom (R4)`;
 - `stone_constructible_card (R1)`;
 - `stone_clopens_card (R2)`;
 - `constructibleTopology_card (R3)`.
- Main theorems `conjecture_false_constructible`, `conjecture_false_clopens` and `conjecture_false_constructibleTopology`. Each has the form

$$\neg \forall (R : \text{Type}) [\text{CommRing } R] (d : \mathbb{N}), \text{ringKrullDim } R = d \rightarrow \#(\text{Stone space}) \leq 2^{2^d},$$
 and is witnessed by $R = \text{Fin } 3 \rightarrow \text{ZMod } 2$.
- Axiom audit: `#print axioms` for the three main theorems, `stone_card_of_hom` and `every_subset` reports `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide` (the ordinary `decide` term is used once, inside `evalAt`, to decide membership $x \in \varphi(b)$).